

Security Decks

Other Resources

[Overall Security Help Page](#)

[Databricks Security Controls](#)

[Product-specific Security Content](#)

[SQRC Knowledge Base](#)

[White Papers](#)

[Questionnaires](#)

Primary Decks (Tier 1)

[Multi-Cloud Pitch Deck](#)

High level information for executives and non-technical security leadership

[Security Overview Deck](#)

Primary resource for security of the overall Databricks Platform. Covers architecture, best practices, customer-configurable security controls and Databricks-managed security controls. **Serverless AND Classic**

Common Decks (Tier 2)

[Serverless Security](#)

Isolation & life cycle for Apps, Custom Models, DBSQL, DLT, Notebooks, Jobs, Workflows

[Custom Model Serving](#)

The security of the life cycle for model serving, including image build

[Mosaic AI Services](#)

MosaicAI services: Vector Search, Online Tables, Foundation Models, FM Training, AI Gateway

[Serverless Appendix](#)

Built-on services like Lakeflow Connect, Lakehouse Monitoring, Predictive Optimization

[Lakehouse Apps](#)

[Default Storage](#)

[Clean Rooms](#)

[Genie Code / Spaces](#)

Tier 3 (and Tier 1+2): [go/platformsecurity/decks](https://go.databricks.com/platformsecurity/decks)

SME: [go/platformsecurity/advanced](https://go.databricks.com/platformsecurity/advanced) (Only available for SMEs and Platform SSAs)

Related Resources

White Paper: [AI Features Security and Governance](#)

Security FAQs:

- [AI Features Security FAQ](#)
- [Relevant Values FAQ](#)

Other FAQs:

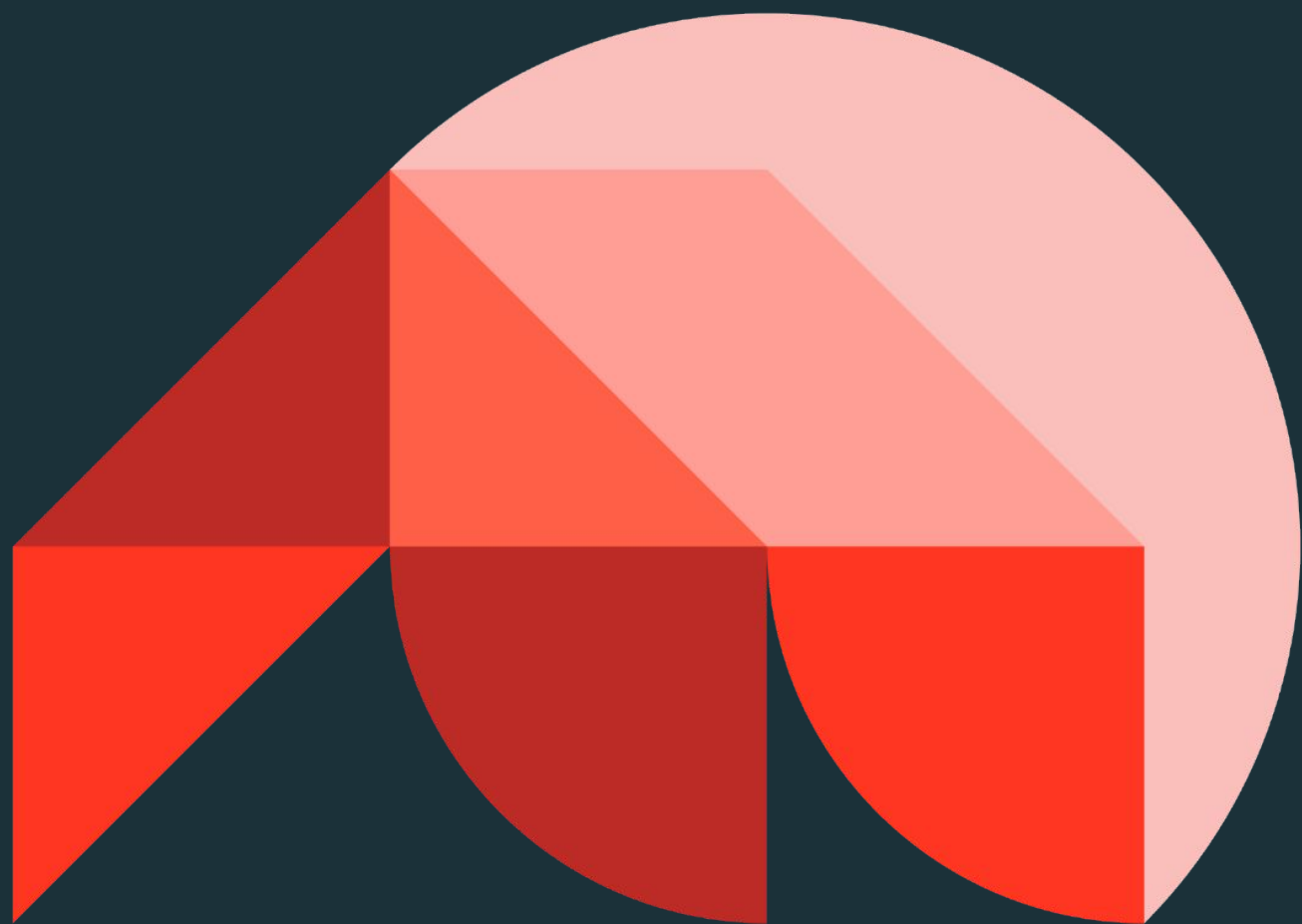
- [Genie Code FAQ](#)
- [Data Science Agent FAQ](#)
- [Genie Research Agent FAQ](#)



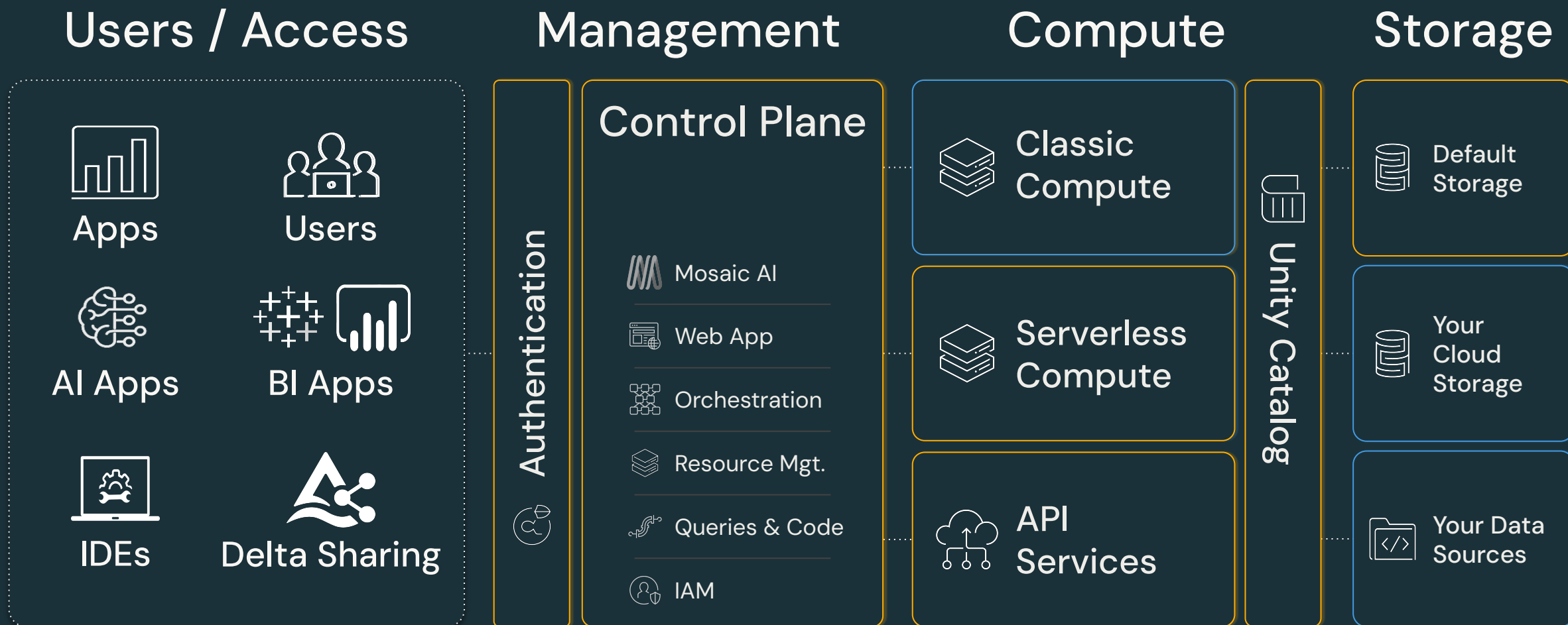
AI Features

Architecture & Security

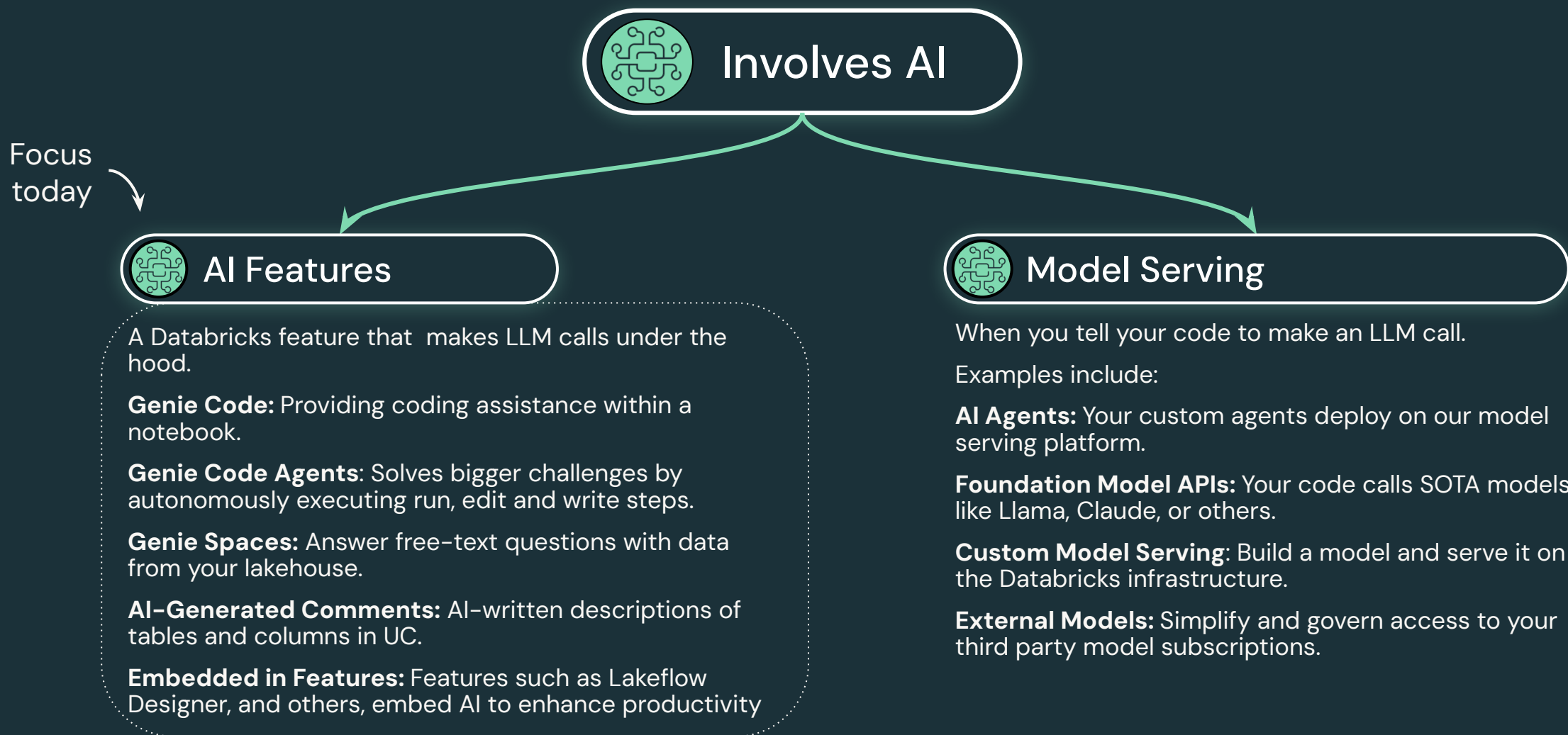
Databricks Data + AI Security



Databricks Platform Architecture



What Are AI Features?



How Do AI Features Operate?

AI Features use an LLM to answer questions, fix code, or automate multi-step tasks using data and metadata from your Lakehouse.



Examples AI Features

A Databricks feature that makes LLM calls under the hood.

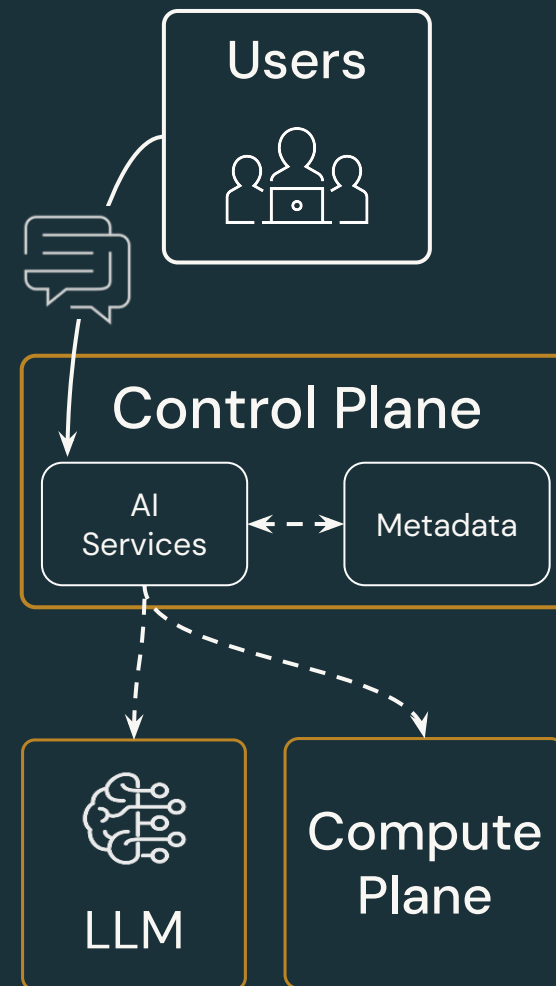
Genie Code Providing coding assistance within a notebook.

Genie Code Agents: Solves bigger challenges by autonomously executing run, edit and write steps.

Genie Spaces: Answer free-text questions with data from your lakehouse.

AI-Generated Comments: AI-written descriptions of tables and columns in UC.

Embedded in Features: Features such as Lakeflow Designer, and others, embed AI to enhance productivity



What Models Are Used?

Whichever models or services are used, the same set of AI-powered features trust and safety commitments ([AWS](#), [Azure](#), [GCP](#)) apply.

Feature	Model Used (May Change)
Genie Spaces	Azure OpenAI service or Anthropic on Databricks
Genie Code Chat	Azure OpenAI service or Databricks-hosted
Genie Code Agents	Azure OpenAI service or Anthropic on Databricks
Genie Code Autocomplete	Databricks-hosted
AI-generated comments	Azure OpenAI service or Databricks-hosted
AI/BI dashboard AI-assisted visualizations	Azure OpenAI service

Databricks continues to evaluate the use of new models and services. See our [documentation](#) for the latest.

What Data is Sent to the Models?

An overview of the data sent to underlying LLMs.

This will vary based on the feature and as they evolve, but generally:

- **User inputs:** Questions or conversations
- **Metadata about the environment:** Table and column names and descriptions
- **Code:** Admin direction for LLMs, sample queries showing how to use tables in practice, and code within notebooks and pipelines
- **Relevant values:** Sampled data from your tables, query results, cell or outputs outputs

Find more detail on our [AI Trust and Safety page](#).

Security and Privacy Controls



AI Feature Security and Governance



Model Partners

- | | |
|---|---|
| Models do not store any data | ✓ |
| Model partners implement zero data retention guarantees | ✓ |
| Model partners guarantee to not train on this data | ✓ |
| Only send data needed for service | ✓ |



Data

- | | |
|------------------------------|---|
| AES-256 encryption | ✓ |
| CMK for managed services | ✓ |
| Genie Space admins can audit | ✓ |



Databricks Governance

- | | |
|---|---|
| Adheres to UC permissions | ✓ |
| Databricks does not train generative models with data submitted | ✓ |
| Public Trust & safety commitments | ✓ |
| Public Responsible AI information | ✓ |



Network

- | | |
|---------------------|---|
| TLS 1.2+ encryption | ✓ |
| Rate-limit controls | ✓ |

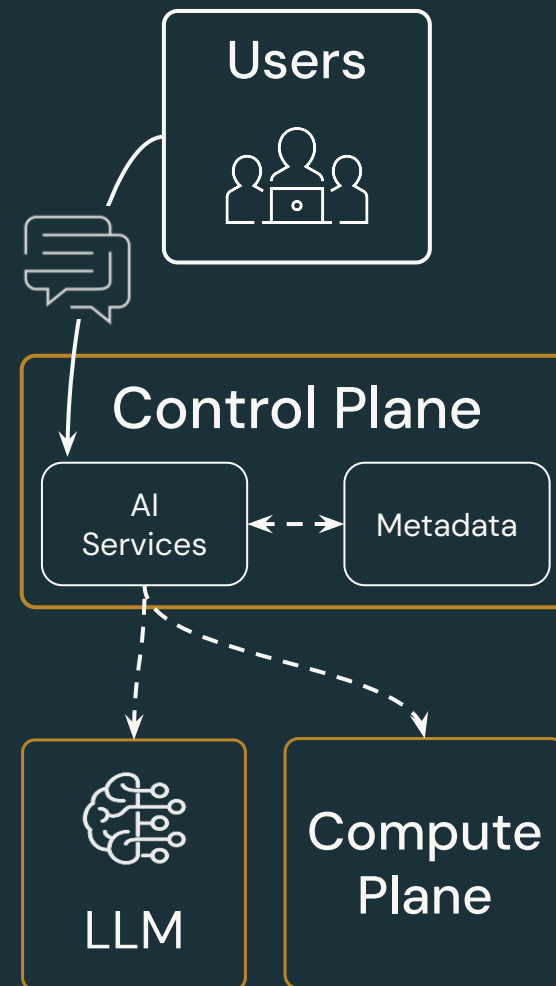


Feature-Specific Details

Genie Spaces

Allows users to ask natural language questions of your data.

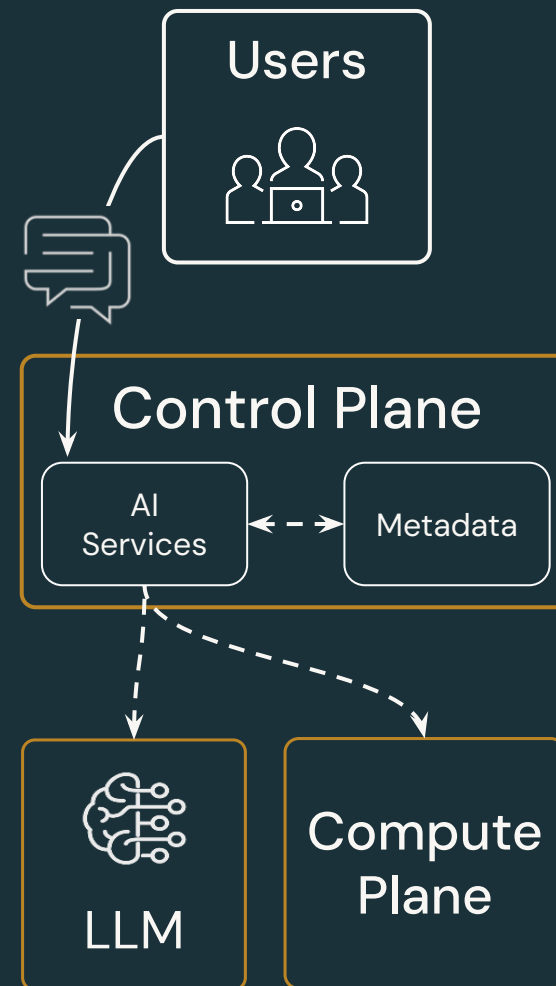
- Experts can curate Genie spaces with instructions, relevant tables, and sample queries
- The **Genie space** sends a user question enriched with metadata and relevant values to an LLM, which typically generates a SQL query
- DBSQL uses the user's identity and queries are read-only
- All UC permissions and controls are enforced
- Users can share conversations, just like queries or notebooks
- **Genie Research Agent** can check results and iteratively re-run queries to improve quality



Genie Code

Embedded in the UI to generate code and debug errors.

- Originated as a simple co-pilot that will send code and metadata from the active cell or notebook and the user: favorite tables, queries
- The **Data Science Agent** can iteratively edit, review and run cells to solve large problems, akin to other code assistants.
- The **Data Engineering Agent** is like the Data Science Agent and helps data engineers create, edit, and debug Lakeflow Declarative Pipelines through natural language interactions



Databricks Security & Trust

Databricks Security Resources



Databricks has accelerators that simplify securing Databricks environments.



Security & Compliance Resources

Due diligence package

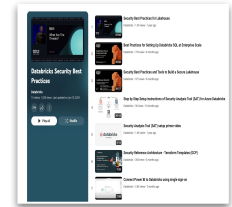
Download

Due Diligence
Compliance Package



Enterprise Security Guide for AWS Deployment

Enterprise
Security Guide



Youtube Best
Practices Videos

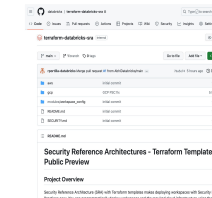


Pre-built Security Tools

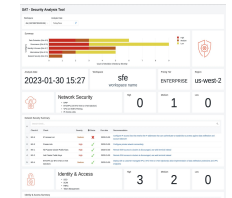
Security Best Practices for Databricks on AWS



Security Best
Practices Guide



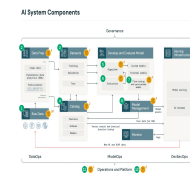
Security Reference
Architecture



Security Analysis
Monitoring Tool



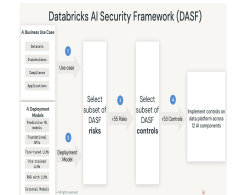
Databricks AI Security Framework



Explains the AI
Lifecycle



Defines AI Risks
and Threats



Defines the Controls
that Protect AI Systems

Appendix

Relevant Values

Why Relevant Values

Genie is a good example, though the same principles apply to other AI features.

- Initially Genie sent metadata and user questions to the LLM model.
- Customer studies showed the model must better understand data. Some users were already pasting in data from tables.
- We added features that will send small sets of relevant values, but not the entire tables, to improve the quality and manage the experience across the platform.
- We continue to work with users and evolve how and when we use relevant values.

Example Sales Table

Make	Model	Amount
Ford	F-150	\$500,000
Ford	Escape	\$350,000

User Query: "Ford F150 Sales all time"

Query broken with only metadata:

```
select sum(amount) from sales
where model='F150'
```

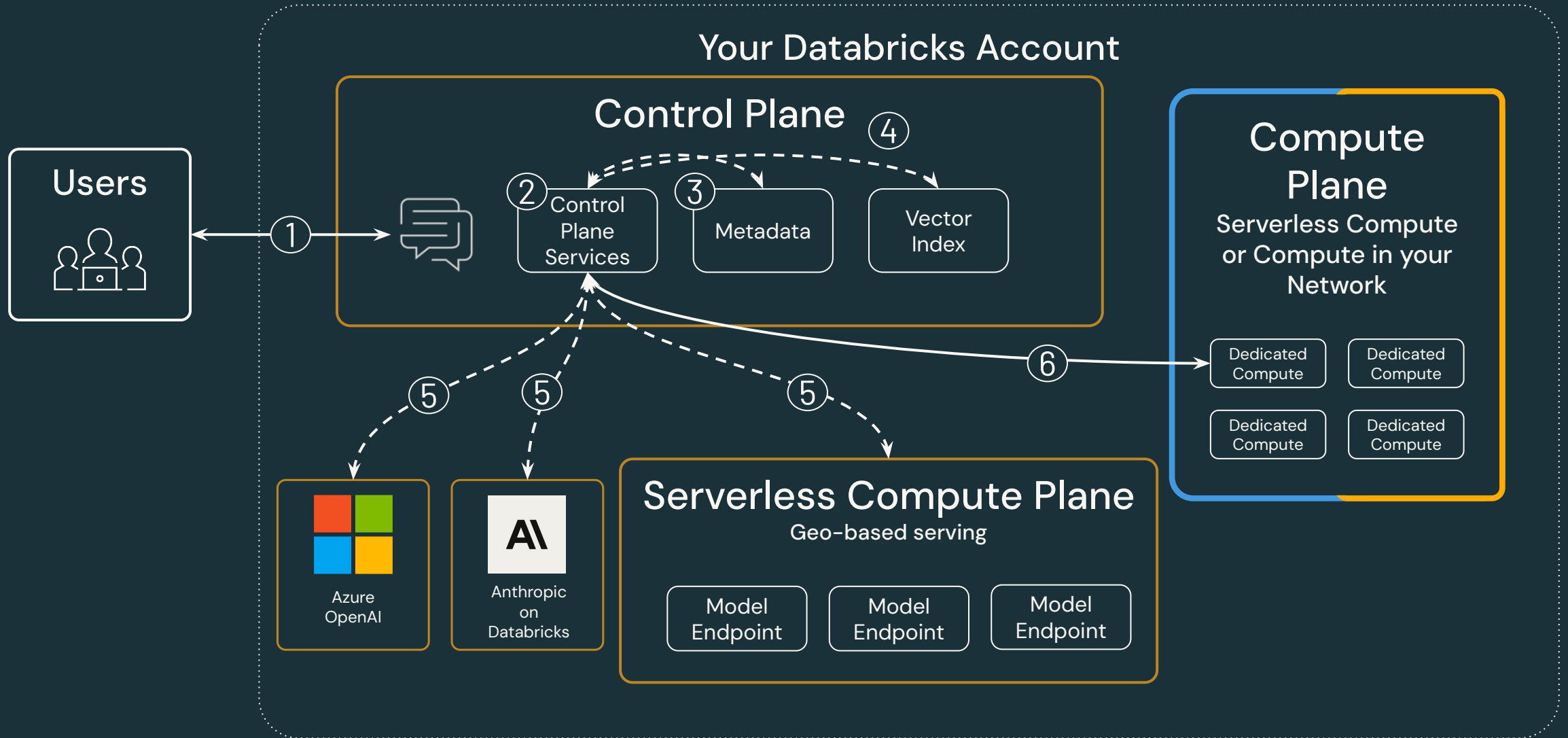
With Data Sampling, query is correct:

```
select sum(amount) from sales
where model='F-150'
```



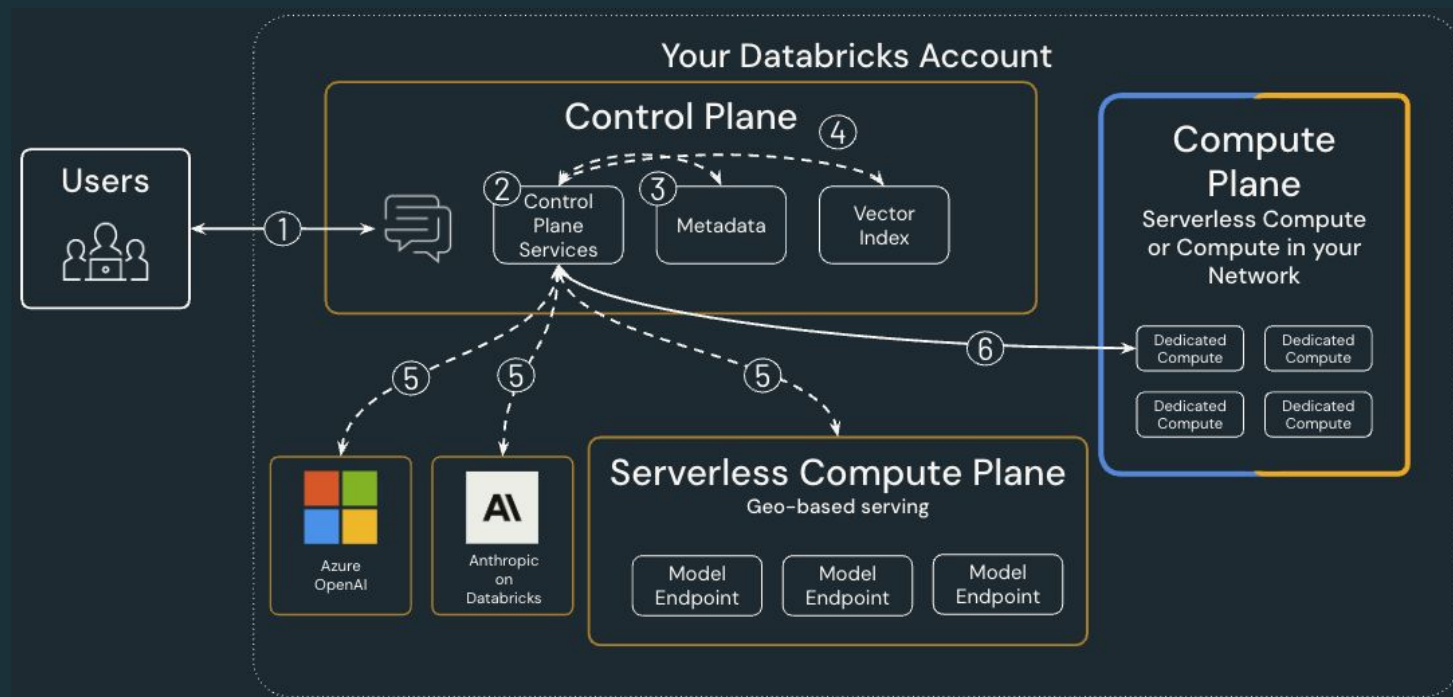
AI-Powered Features General Architecture

AI-Powered Features – General Architecture



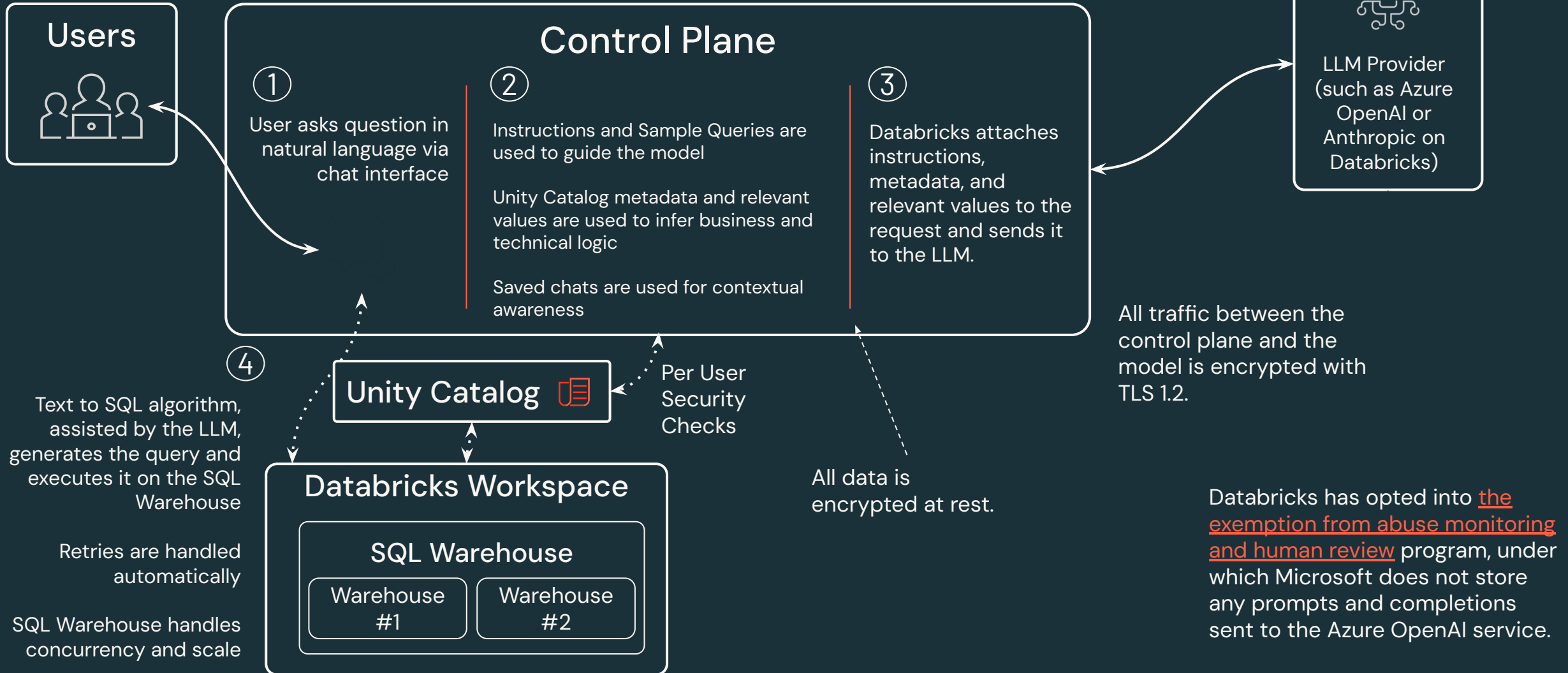
AI-Powered Features – General Architecture

1. Users connect to Databricks and authenticate and authorized via normal mechanisms.
2. AI assistive features run as isolated control plane services, AI assistive features use the standard control plane service security mode
3. User interactions attach metadata and relevant values that respect Unity Catalog permissions
4. Metadata is encrypted at rest (AES-256) and protected by customer-managed keys when configured
5. Databricks sends metadata, relevant values, and user input to Azure OpenAI, Anthropic on Databricks, or Databricks-managed models
6. Users optionally accept/run AI suggestions and decide whether to execute generated code



Genie Spaces General Architecture

Genie Spaces – General Architecture

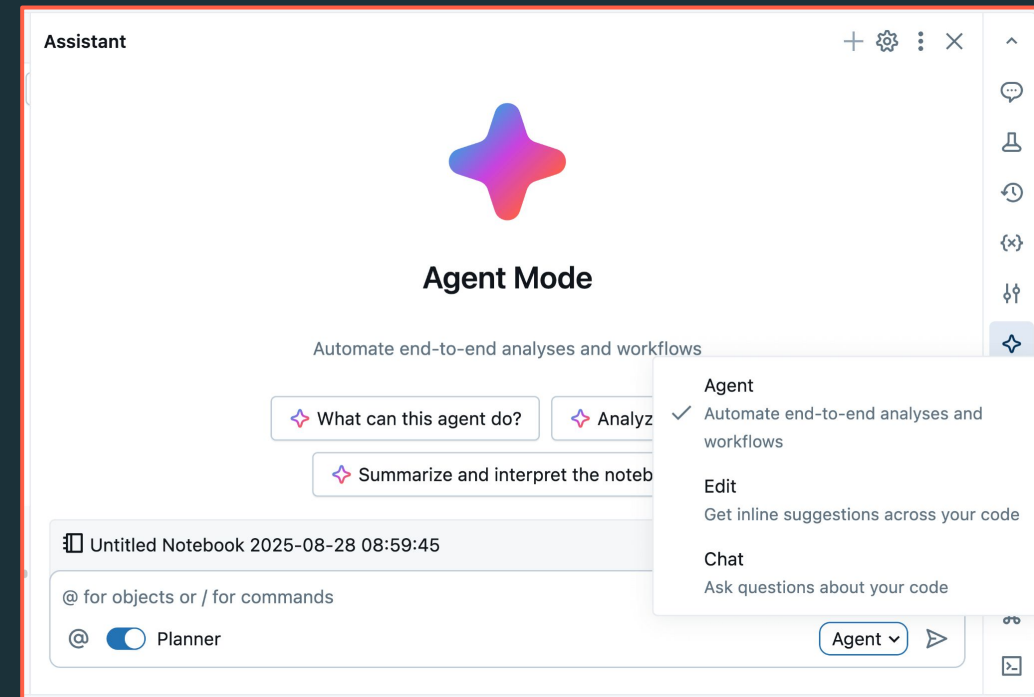


Genie Code Agents Overview

Genie Code Agents

Overview:

- **Data Science Agent:** The agent can automate multiple steps. From a single prompt, it can retrieve relevant assets, generate and run code, fix errors automatically, and visualize results. It adds the ability to sample data and cell outputs to provide better results.
- **Data Engineering Agent:** The Data Engineering Agent is an AI-powered assistant that extends the existing Data Science Agent to help data engineers create, edit, and debug Lakeflow Declarative Pipelines through natural language interactions. The agent operates within the Databricks workspace UI, specifically in the Lakeflow editor, enabling users to describe high-level tasks that the agent then implements by updating pipeline files, running validations, and iteratively debugging errors.

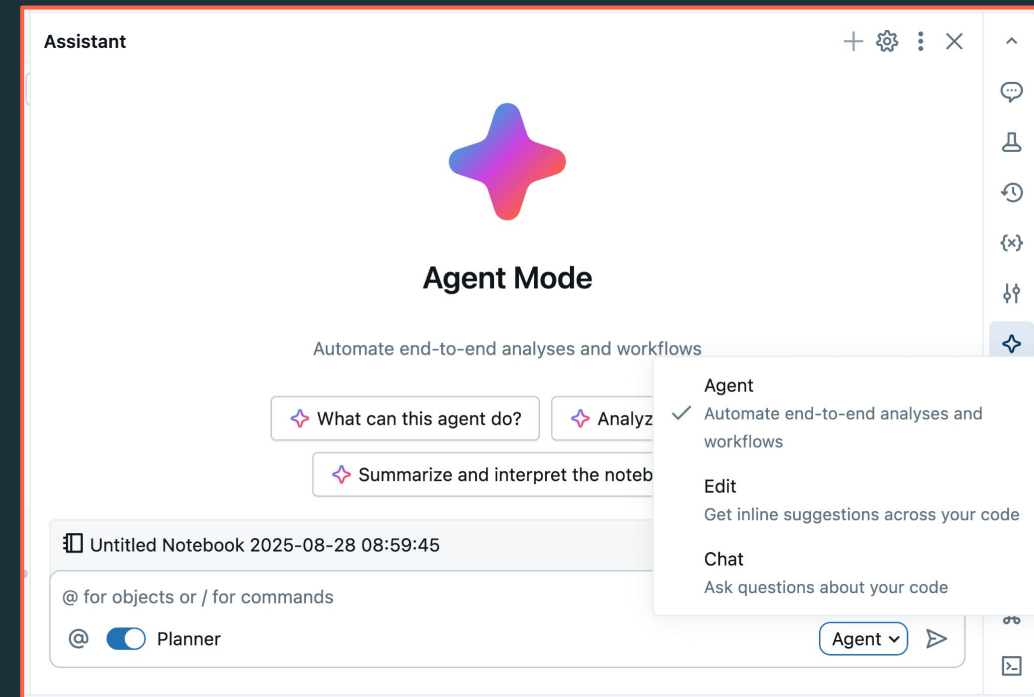


Beta: Data Science Agent

Genie Code Agent Mode, Data Science Agent, beta will soon be **enabled by default**. (Excludes workspaces where with the Compliance Security Profile is enabled)

Details:

- Genie Code in Agent Mode will choose between **Azure OpenAI** or **Anthropic on Databricks** (uses endpoints hosted by Databricks Inc. in AWS within the Databricks security perimeter), and is only available when the partner-powered AI features setting is enabled.
- Admins can disable the beta if needed until the feature reaches General Availability.



General AI Features

FAQ

General FAQ Continued

What data is sent to AI features?

We send only what's needed to serve the specific user request and stay within token limits. Think narrow slices: representative values for matching, prior results to enable or compact error/context snippets—not bulk exports.

Is any content filtering applied to filter for harmful responses?

When using Azure Open AI, Databricks also uses Azure OpenAI [content filtering](#) to protect users from harmful content. In addition, Databricks has performed an extensive evaluation with thousands of simulated user interactions to ensure that the protections put in place to protect against harmful content, jailbreaks, insecure code generation, and use of third-party copyright content are effective.



General FAQ

What models are you using?

Answer: Please refer to diagram from previous slide.

What models are used?

Whichever models or services are used, the same set of AI-powered features trust and safety commitments ([AWS](#), [Azure](#), [GCP](#)) apply.

Feature	Model Used
Databricks Assistant	Azure OpenAI service or Databricks-hosted
Databricks Assistant Data Science Agent	Azure OpenAI service or Amazon Bedrock
Databricks Assistant Autocomplete	Databricks-hosted model
AI-generated comments	Azure OpenAI service or Databricks-Hosted
AI/BI dashboard AI-assisted visualizations	Azure OpenAI service
Genie spaces	Azure OpenAI service

Is my data being used to train models?

Answer: No, neither Databricks nor our model partner (Azure OpenAI or Anthropic on Databricks) trains generative models using customer data.

Is row-level data being sent to the models?

Answer: Generally, these features will send small amounts of data from your tables or results to the model to improve accuracy and functionality, like similar tools in the industry.

Examples:

- [Genie data sampling](#) improves accuracy by allowing Genie to use sample data, helping it align user text to the values and structure of your data.
- Genie Code [Agent Mode](#) analyzes cell outputs and read data samples from tables.



General FAQ Continued

Do Azure OpenAI or other model partners collect my data?

No. Databricks has opted into the exemption from abuse monitoring and human review program, under which Microsoft does not store any prompts and completions sent to the Azure OpenAI service.

Are there any data residency considerations I should be aware of?

Summary:

- AI features are enabled with Databricks [Geos](#).
- Azure Databricks AI features are included within the EU Data Boundary.
- Workspace admins can optionally enable or disable data processing outside of their Geo.

What data is encrypted by default?

Summary:

- AI traffic between the control plane and the Azure OpenAI service or other model partners ($\geq$ TLS 1.2)
- All traffic between the control plane and Databricks-managed models ($\geq$ TLS 1.2)
- All Databricks AI features related storage (AES 256 bit)
- Optionally enable customer-managed keys (CMK) covering Genie Code and Genie Spaces.





Genie Code Agents FAQ

Genie Code Agents FAQ

What data is being sent to the LLM?

In addition to the user's working context used by Genie Code, the Agents generate responses anchored in your organizational data. This can include using data samples from tables or cell outputs to enable the agent to provide relevant suggestions.

Are there any guardrails to prevent dangerous code execution?

Genie Code Agents have built-in LLM guardrails when using the code execution tool to help reduce unintended actions, such as destruction of data (e.g., tables/files/schemas), changing system permissions, calls to external websites, or other malicious impact. That said, we still recommend reviewing generated code carefully, especially when it touches production data, important tables, or other sensitive operations.



Genie Code Agents FAQ Continued

Do all Genie Code Agents have the same security controls?

Yes, all Genie Code Agents, such as the Data Science Agent and the Databricks Engineering Agent have the same underlying pattern, but purpose-built for different use cases.

How can customer teams or customer security teams monitor the behavior Genie Code Agents?

The Genie Code Agents behavior will appear in audit logs as the user's actions. If your customer has requirements on monitoring Genie Code Agents specifically, please let us know so we can gather more information.



Genie Code Agents FAQ Continued

Are the Genie Code tools usage limited based on the user's permissions?

Yes, any Genie Code Agents can only read data samples, read cell outputs, or execute cells where the user is able to run that operation.

Can customer teams set limitations about the types of actions that their users can perform with the agent?

Not today. If you have this requirements about this, please let us know.



The background is a dark blue-grey color. It features several abstract geometric shapes in a vibrant orange-red and a muted teal. In the top right, there is a large orange circle. Below it, towards the center-right, is a smaller orange square. Further down and to the right is a teal triangle pointing upwards. In the bottom right corner, there is a large orange triangle pointing downwards, and a small orange circle is positioned just above it. The text "Thank you!" is written in a large, white, sans-serif font on the left side of the slide.

Thank you!